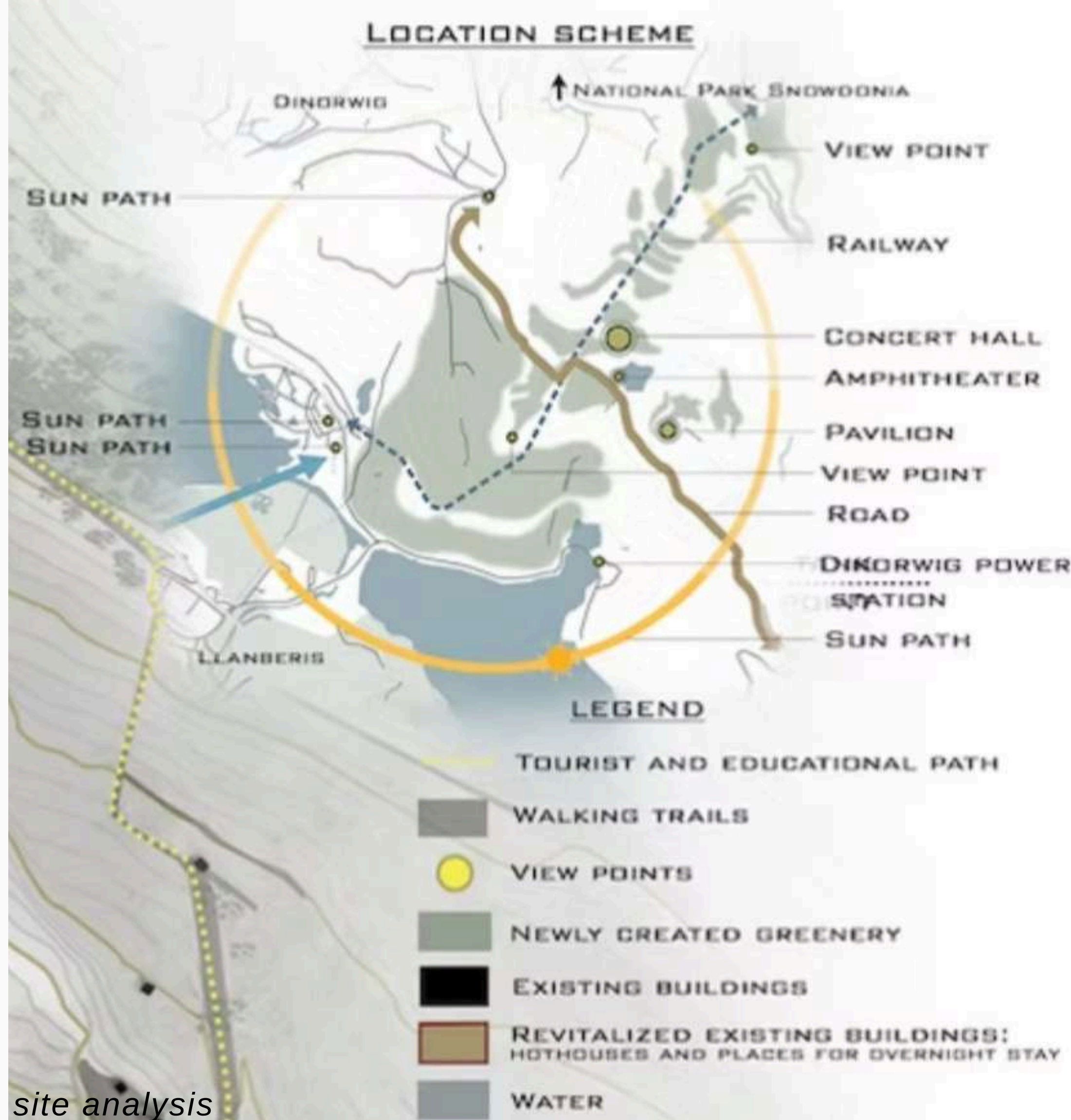
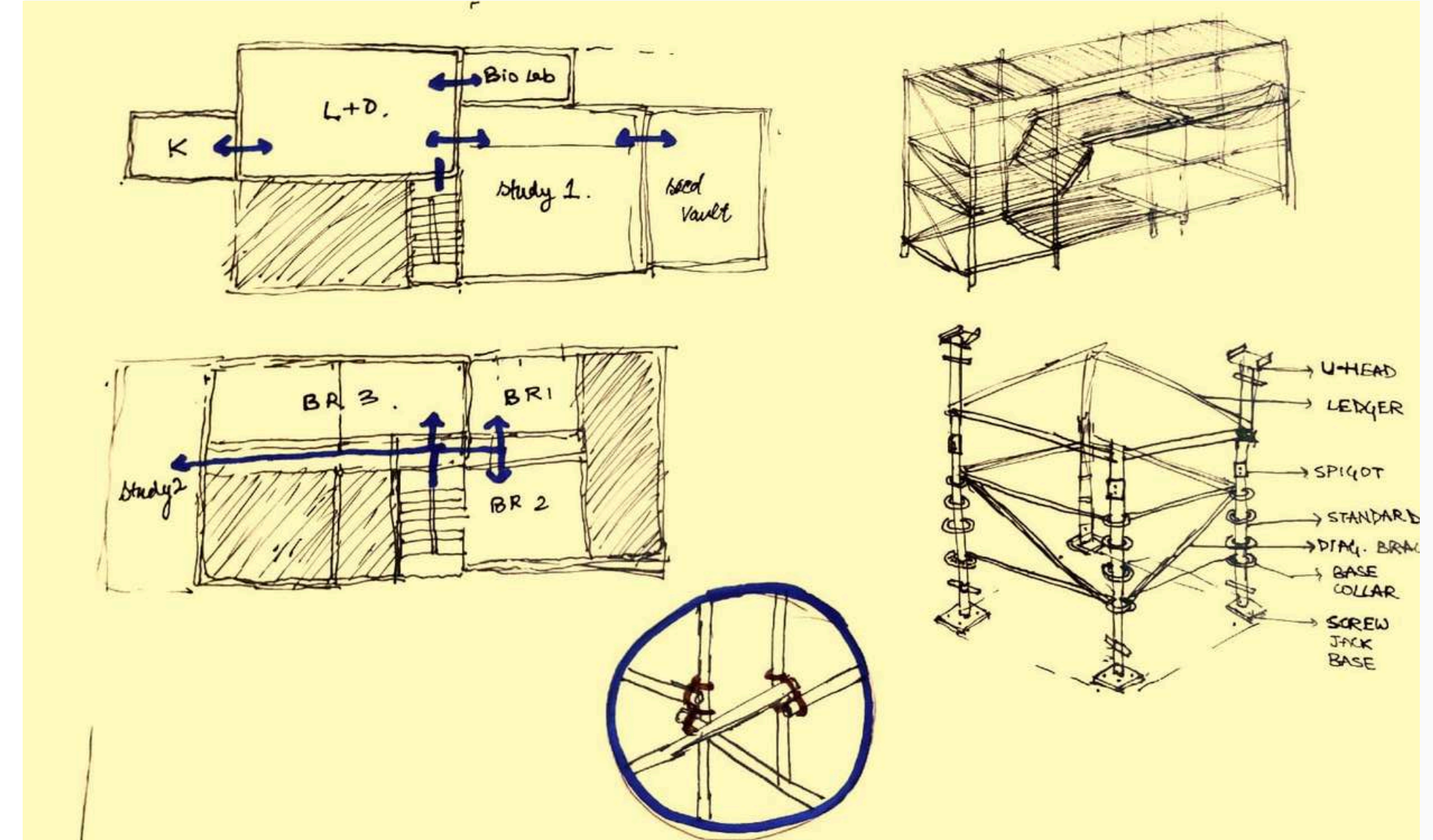




site analysis

Site & Context Analysis

Dinorwig is a historic slate quarry in Gwynedd, North Wales, near Snowdonia. Strong air currents moving through the gorge also form natural flight paths for local birds. The site is surrounded by native wildlife, including Sessile Oak, Scots Pine, Lady Fern, and Mountain Buckler. To protect this rewilding landscape, the home uses a lightweight frame anchored directly into rock fissures, leaving the quarry floor completely untouched.

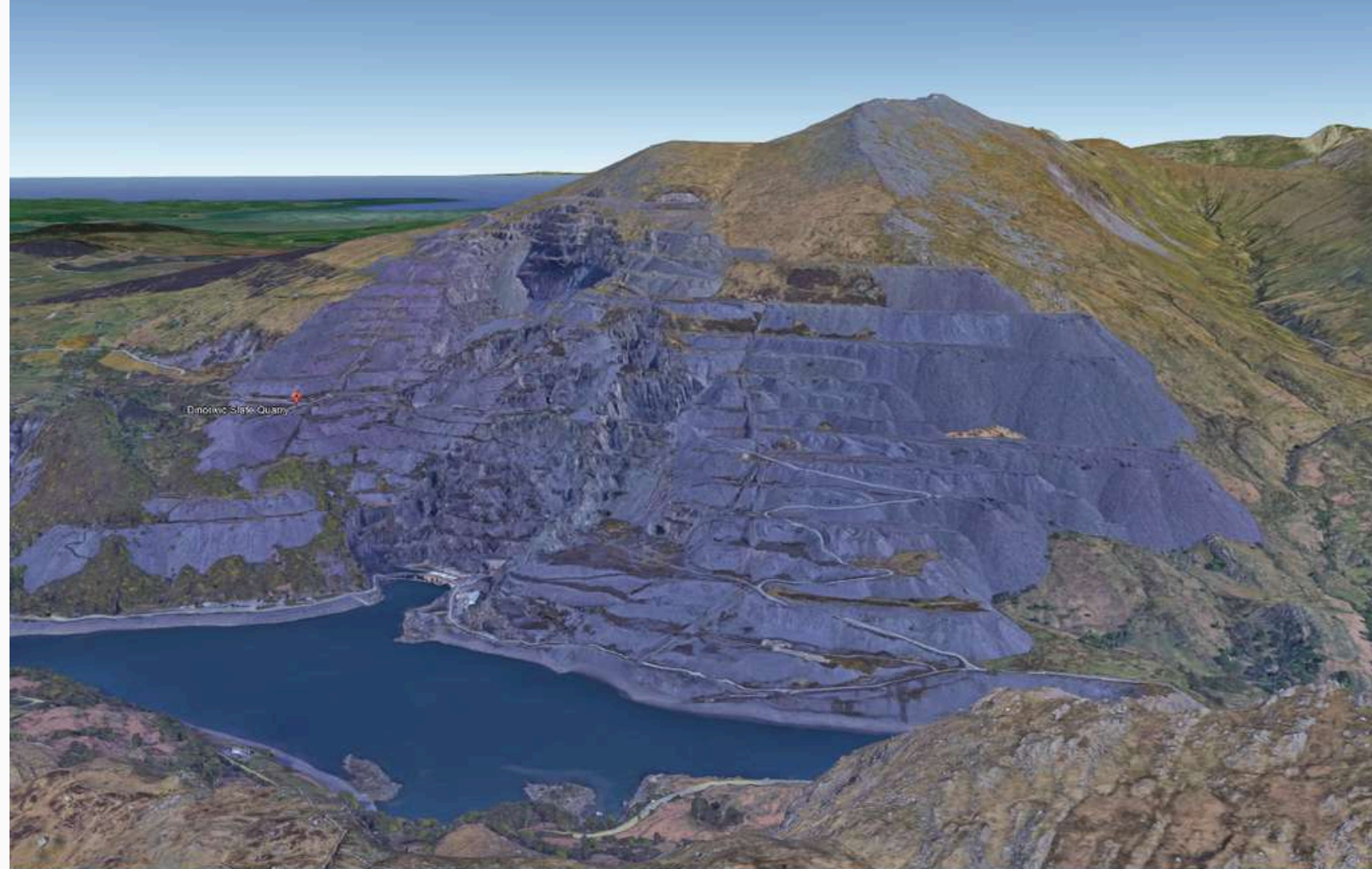


concept developement

Concept & Vision

Parasitic Integration: Lightweight modular pods suspended directly off a sheer slate cliff face, minimizing ground footprint while maximizing exposure to the quarry gorge ecosystem.

Symbiotic Program: Blends living quarters with specialized mycological research (bio-lab and spore vault) and flora/fauna interaction.



2026

- *Initial vegetation is sparse—mostly bare slate terraces with emerging mosses along moist crevices.*



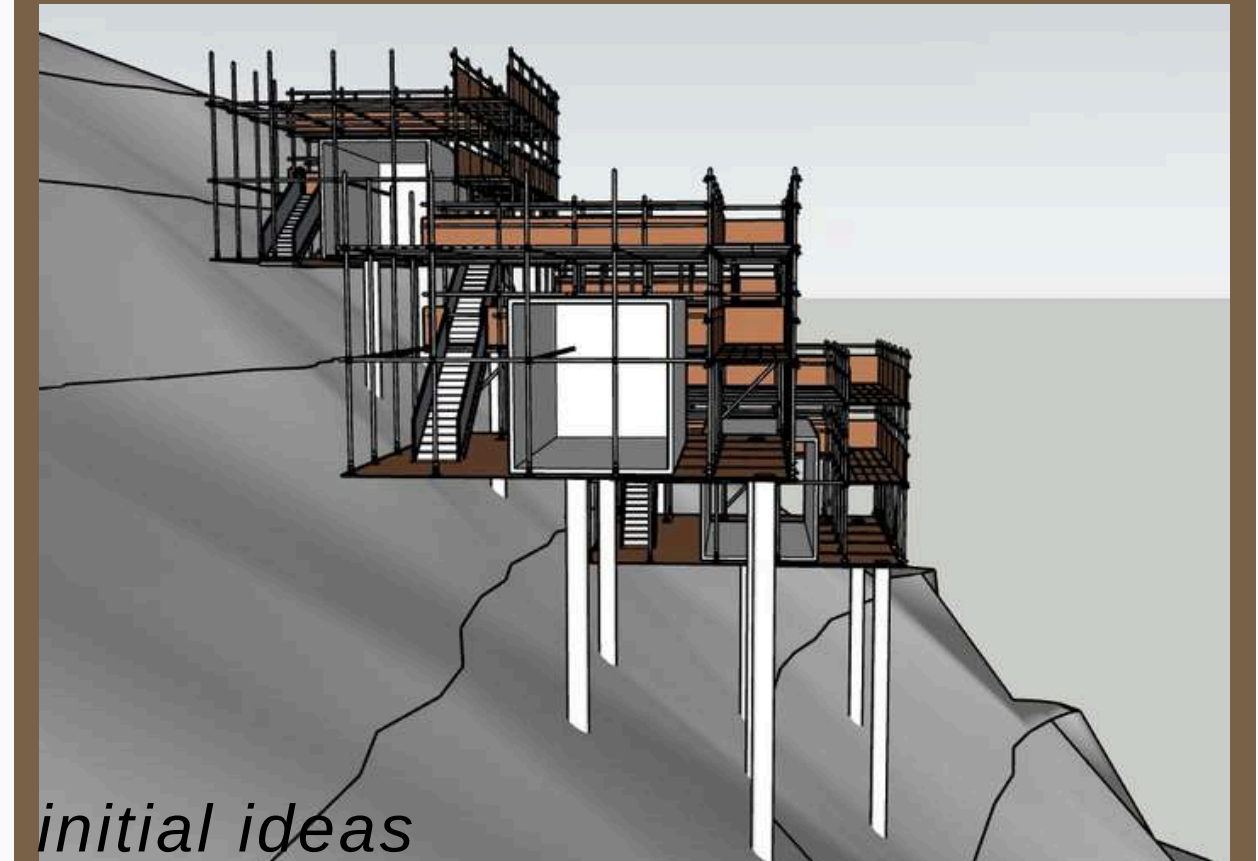
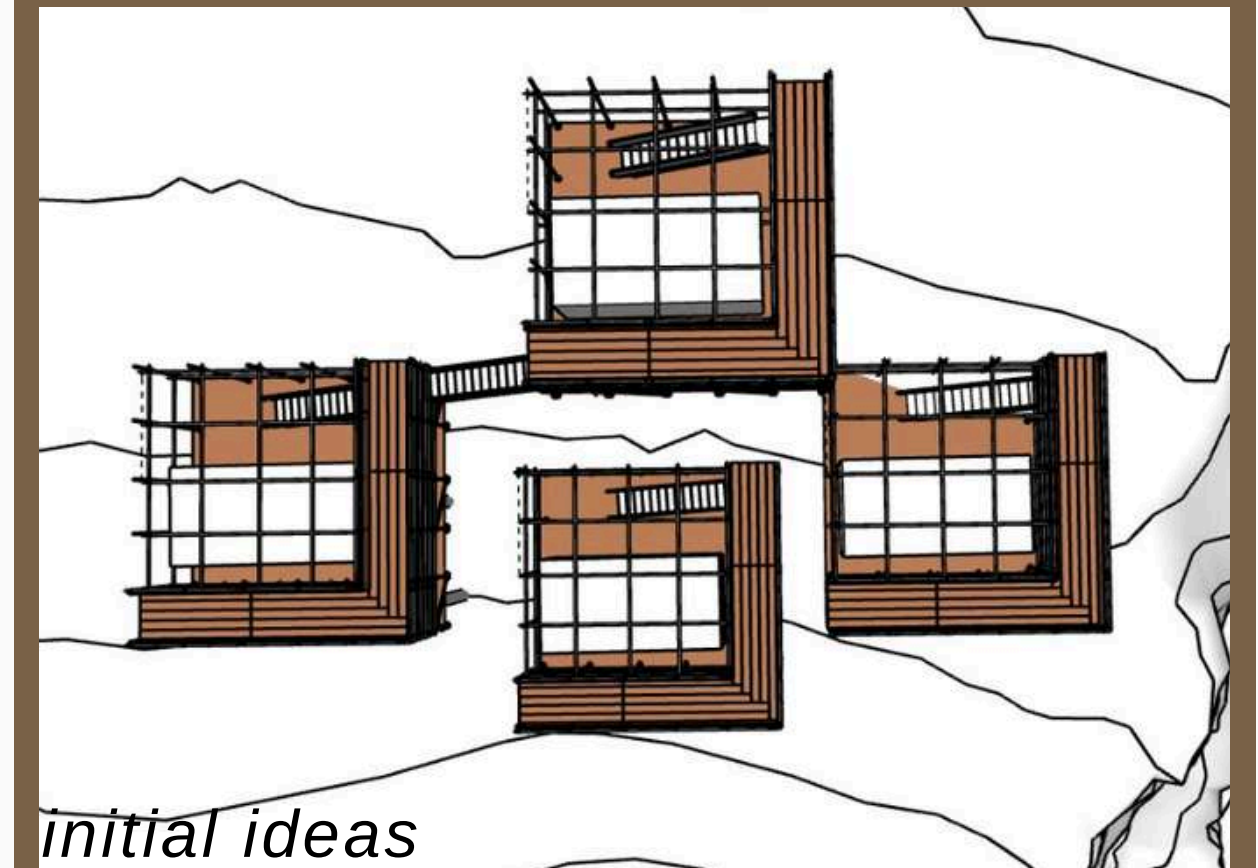
2045

- *Pioneering moss, mountain ferns, and young rowan trees take root in wet rock fissures.*



2060

- *Purple heather, yellow gorse, and dense Scots pine pockets cover the lower and middle terraces.*
- *Top cliffs remain stark and wind-swept, surrounded by frequent mountain fog and lake cloud inversions.*



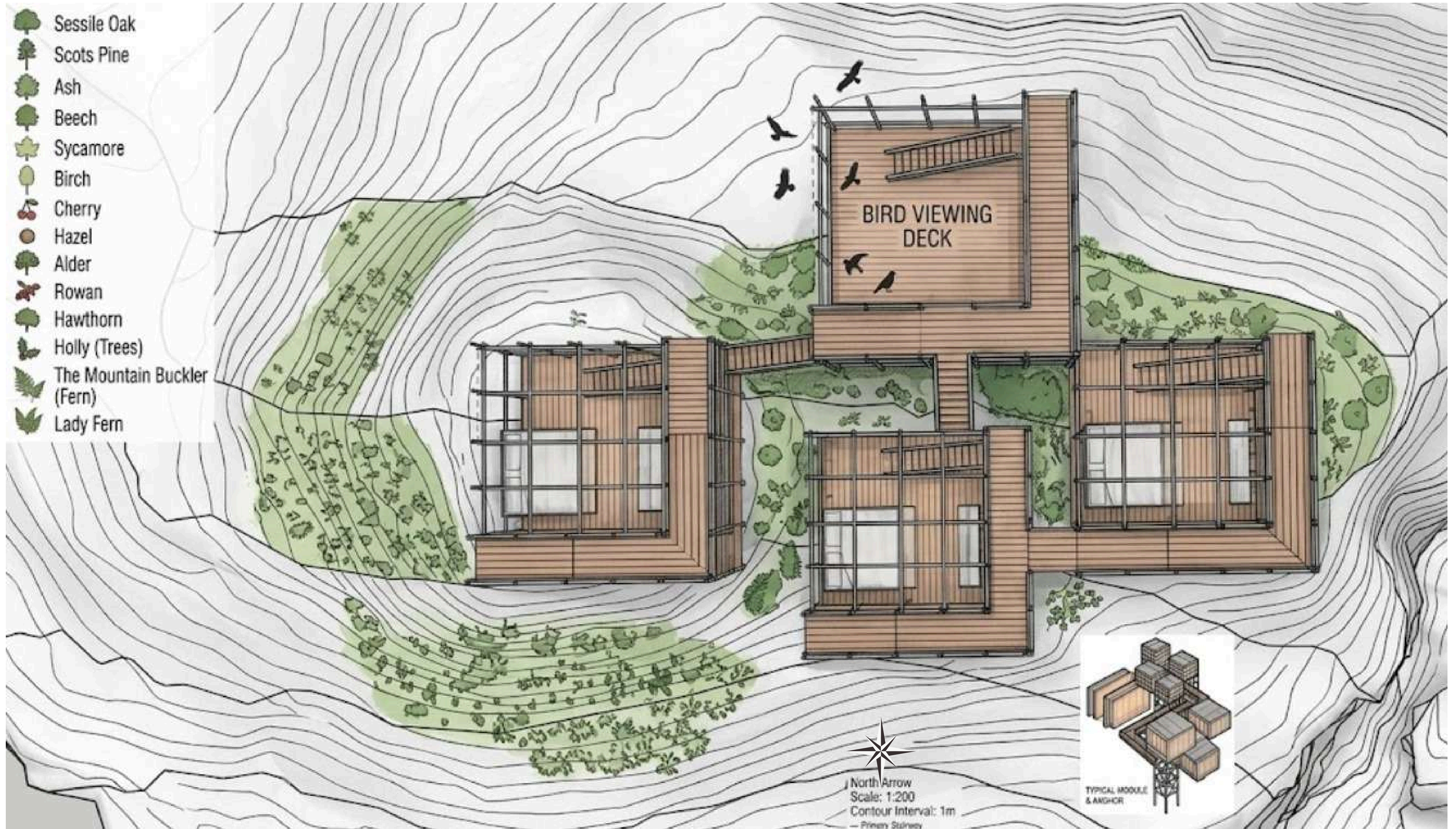
Structural & Material Matrix

Primary Frame: High-tension 80mm modular galvanized steel tubular framework, rock-pinned via 32mm stainless steel anchors with 1.2m cliff offset.

Thermal Envelope: Double-wall 16mm multi-layer translucent polycarbonate pods with passive trapped-air thermal buffers.

Circulation Grid: Step-connected timber and perforated steel catwalks bridging all four stepped living modules.

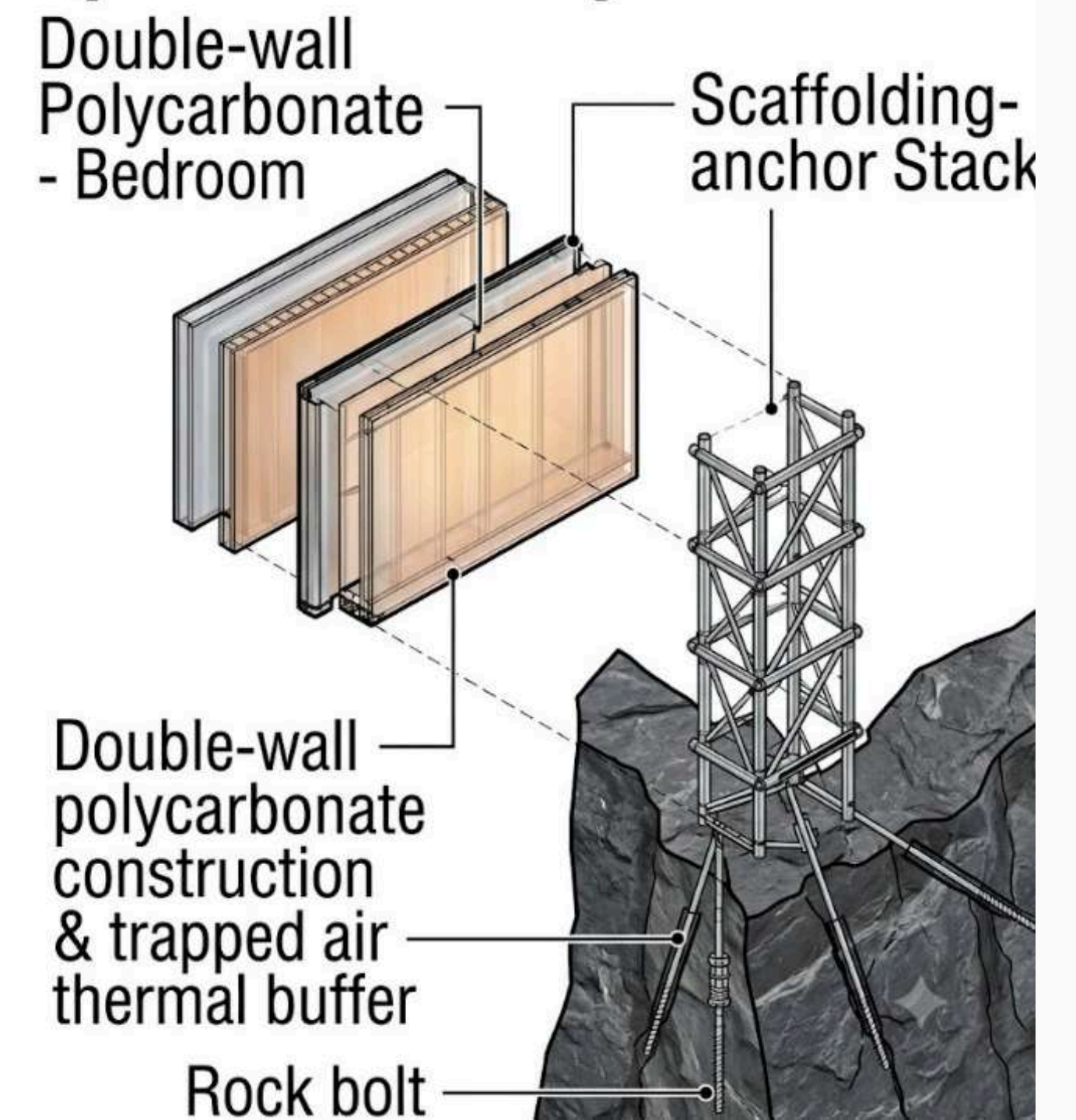
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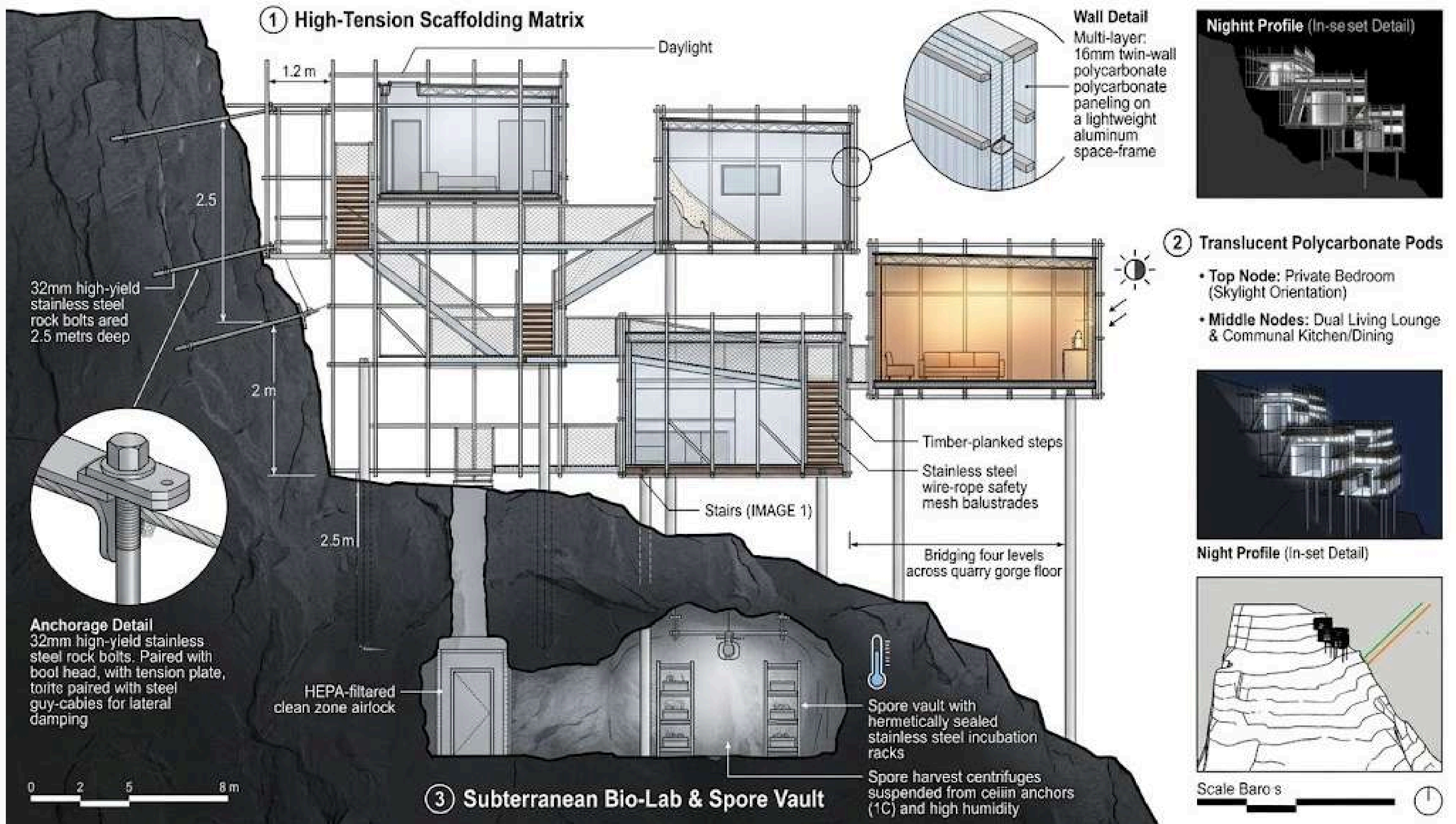
Plan mapping the four modules joined by full stairways. Highlights the topmost Bird Viewing Deck and the surrounding contour zone rewilded with mature native flora (Sessile Oak, Scots Pine, Lady Fern, Mountain Buckler, etc.).



Exploded Assembly



Front profile displaying the 4 stepped modules, translucent thermal skins, and perforated catwalks against the weathered slate cliff. Showcases the site's light-touch footprint and full structural integration with rising gorge air currents.



Vertical cut section revealing the scaffold framework, 1.2m cliff offset, and double-skin polycarbonate pods suspended above the quarry. The scaffold terminates in the subterranean slate cave, housing the bio-lab and spore vault at natural thermal equilibrium.



Design Synthesis: 2060 Paradigm

Architecture as Parasite & Host: The design rejects intrusive ground foundations in favor of an elevated 80mm galvanized frame anchored into slate fissures. The structure behaves as a non-invasive visitor clinging to the vertical rock face.

Passive Bio-Integration: Polycarbonate pods and the slate cave bio-lab rely entirely on natural thermal mass and microclimate buffers, maintaining zero-energy climate control for both human habitation and mycological research.

Rewilded Landscape Matrix: By 2060, the contour-planted native flora (Sessile Oak, Scots Pine, Lady Fern) has fully matured around the frame, seamlessly merging avian flight paths, structural catwalks, and fungal spore cycles into a single coexisting ecosystem.